Transient Freezing Governs Flat-Minimum Selection in Stochastic Gradient Descent

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Stochastic gradient descent (SGD) often converges to flatter minima with better generalization, but the dynamical event that fixes the final basin remains unclear. Existing theories mainly describe a late-stage or fixed-landscape preference induced by anisotropic noise. Here we show that basin selection is controlled by an earlier nonequilibrium process. In neural-network experiments, SGD first remains mobile among distinct solution valleys and then loses this mobility at a well-defined freezing time. Continuation-training measurements identify the valley occupied by the trajectory at different checkpoints and show that the final basin is selected when inter-valley motion arrests. Increasing SGD noise delays this freezing time and thereby increases the probability of ending in a flatter, better-generalizing valley. A minimal two-valley model with landscape-dependent noise reproduces the observed hopping, delayed freezing, and final selection. The model also resolves an apparent paradox: stronger noise can weaken the instantaneous fixed-slice preference for the flat valley while enhancing the final flat-minimum probability by shifting the freezing point. These results recast the implicit bias of SGD as a history-dependent nonequilibrium selection mechanism controlled by the duration of transient mobility.

I. INTRODUCTION

Deep learning works by moving through a high-dimensional loss landscape, but the principle that selects the final solution is still not fully understood [1–4]. A robust empirical observation is that stochastic gradient descent (SGD) tends to find flatter minima than less noisy optimization procedures, and such minima are often associated with better generalization [5–10]. The useful question is therefore not only why flat minima are preferred, but when this preference becomes an irreversible basin choice.

Most existing physical pictures address the first part of this question. They show that minibatch noise is anisotropic and correlated with the local Hessian, so SGD does not sample the loss landscape as an equilibrium thermal bath would [11–14]. This geometry-coupled noise can generate an effective preference for flatter regions. Such arguments are powerful, but they are most naturally formulated in a late-stage or fixed-landscape regime, after the trajectory has already entered a low-gradient valley. They do not by themselves determine how the final basin is chosen during early training, when gradients are large, barriers are changing, and the trajectory may still move among competing valleys.

This distinction matters because several important properties of trained networks are established early in optimization [15–19]. If solution identity is fixed during this transient stage, then a theory based only on asymptotic sampling misses the decisive step. The key variable should be the time at which inter-valley mobility is lost. Before that time, noise can redirect the trajectory among basins; after that time, later training mainly relaxes within the basin already selected.

Here we study flat-minimum selection as a transient nonequilibrium freezing process. The minimal ingredients are competing valleys, landscape-coupled noise, barriers that grow during descent, and a finite window of inter-valley mobility. We show empirically that SGD hops among solution valleys before committing to one basin, define a freezing time t_f that marks this commitment, and demonstrate that stronger noise promotes flatter minima mainly by delaying t_f. We then construct a two-valley model that separates the fixed-landscape bias from the stopping-time effect. The resulting mechanism is simple: SGD selects flat minima not just because flat valleys are statically favored, but because noise keeps the dynamics mobile until a later point along the evolving landscape, where the frozen-in bias is stronger.

II. EARLY-TIME VALLEY HOPPING PRECEDES BASIN COMMITMENT

We first establish that the relevant landscape contains multiple competing basins. Starting from the same initialization, independent SGD runs on the MNIST classification task converge to distinct clusters in parameter space while reaching zero training error. Pairwise comparison of the final classifiers using the Jaccard similarity of their test-error sets shows that these clusters correspond to functionally distinct solutions. Linear interpolation between dissimilar endpoints reveals finite loss barriers, whereas highly similar endpoints are connected by nearly barrier-free paths. Thus the final solutions are not merely different points on one connected floor; they

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occupy distinct valleys separated by barriers.

The more direct test is dynamical. We save a reference SGD trajectory at checkpoint times t_c and restart training from each checkpoint using deterministic full-batch gradient descent. Because this continuation dynamics has little stochastic hopping, the endpoint of the continuation identifies the valley that the stochastic trajectory would occupy if exploration were stopped at t_c. Early checkpoints lead to different continuation endpoints, with different test loss and flatness. Late checkpoints lead to the same endpoint. The valley identity inferred from the continuation therefore changes during early training and then becomes stable.

This protocol gives an operational definition of basin commitment. Let $\mathcal{C}[\theta(t_c)]$ be the endpoint obtained by deterministically continuing the checkpoint $\theta(t_c)$, and let $b(t_c)$ be the basin label assigned to that endpoint by clustering continuation endpoints using functional similarity. For an ordered set of checkpoints $\{t_i\}$, we define

$$t_f \equiv \min\{t_i \mid b(t_j) = b_* \text{ for all } j \geq i\}, \tag{1}$$

where b_* is the stable late-time basin label. This definition is intentionally dynamical. It is not simply the time at which the training loss becomes small; it is the earliest checkpoint after which stopping the stochastic exploration and relaxing deterministically always returns the same basin.

The conclusion is that the decisive event in SGD is the loss of mobility across valleys. Before t_f, the trajectory still samples different basins and can escape sharper regions. After t_f, subsequent optimization mainly refines the solution inside one basin. The origin of flat-minimum selection should therefore be sought in the transient regime before basin commitment.

III. BASIN SELECTION IS SET AT THE FREEZING POINT

The definition of t_f in Eq. (1) identifies when a trajectory becomes committed, but by itself it does not explain why SGD noise changes the final basin distribution. We therefore next ask whether the freezing point is the stopping-time variable through which stochasticity becomes a basin choice. If freezing were merely a post hoc diagnostic, changing the learning rate or batch size could alter solution geometry independently of the measured commitment time. Instead, we find that the dominant effect of stronger SGD noise is to shift the commitment coordinate itself.

For fixed architecture and data, increasing the learning rate or decreasing the batch size increases the effective stochasticity of the update. We measure t_f using the continuation-based basin-stability criterion: early checkpoints continue to endpoints belonging to different basin labels, whereas checkpoints after t_f all relax to the same label. Across trajectories that are continuation-stable and converged at the final checkpoint, the rescaled freezing coordinate ηt_f increases with the effective noise proxy η/B. Stronger stochasticity therefore keeps trajectories mobile across competing valleys for a longer portion of training.

A subtle point is where to measure geometry. With cross-entropy loss, training can continue to decrease the training loss after classification accuracy has saturated, while final test loss and local sharpness may drift within the already selected basin. To avoid conflating basin selection with this post-freezing margin relaxation, we evaluate local geometry at the freezing checkpoint itself. This quantity is not meant to remove all dependence on training age: a later freezing point is also a later point along the optimization trajectory. Rather, it asks what geometric state the trajectory has reached at the moment when the basin identity becomes fixed. We define the freezing-time flatness

$$F_f \equiv -\log_{10} S_{\mathrm{nw}}(\theta_{t_f}), \tag{2}$$

where S_{nw} is a neuron-wise relative sharpness: for each weight matrix, random perturbations are normalized separately for each output neuron, so that the perturbation norm of each incoming weight vector is a fixed fraction of its weight norm. We then average the positive train-loss increase over random directions. This fully connected analogue of filter-normalized perturbations provides a scale-aware probe of the geometry present when basin commitment is made, not a time-independent causal estimate of final-basin flatness or the fully relaxed geometry at the end of training.

This freezing-time geometry is organized more directly by the freezing coordinate than by the raw hyperparameter noise proxy. Conditions with different learning rates and batch sizes scatter when F_f is plotted against η/B, but collapse more clearly when plotted against $\langle \eta t_f \rangle$ for the final-converged stable trajectories. The point of this comparison is not that time alone causes flatness to increase, but that the continuation-defined commitment time provides a better dynamical coordinate for the geometry observed at commitment than the nominal noise scale does. Final test accuracy shows the same qualitative ordering: it is weakly organized by the raw noise proxy and more coherently organized by the freezing coordinate. Thus the relevant role of noise is not only to make the final iterate sharper or flatter; it controls when the trajectory loses inter-valley mobility, and the geometric state present at that arrest point is what subsequent within-basin relaxation inherits.

This statement is stronger than saying that more noise produces more exploration. Exploration matters only while valley exchange is still possible. Two training protocols with different microscopic noise statistics can lead to similar outcomes if they produce similar freezing times, whereas two protocols with comparable nominal noise can differ if one freezes much earlier. The physically meaningful quantity is therefore not the raw hyperparameter alone, but the point along the evolving landscape

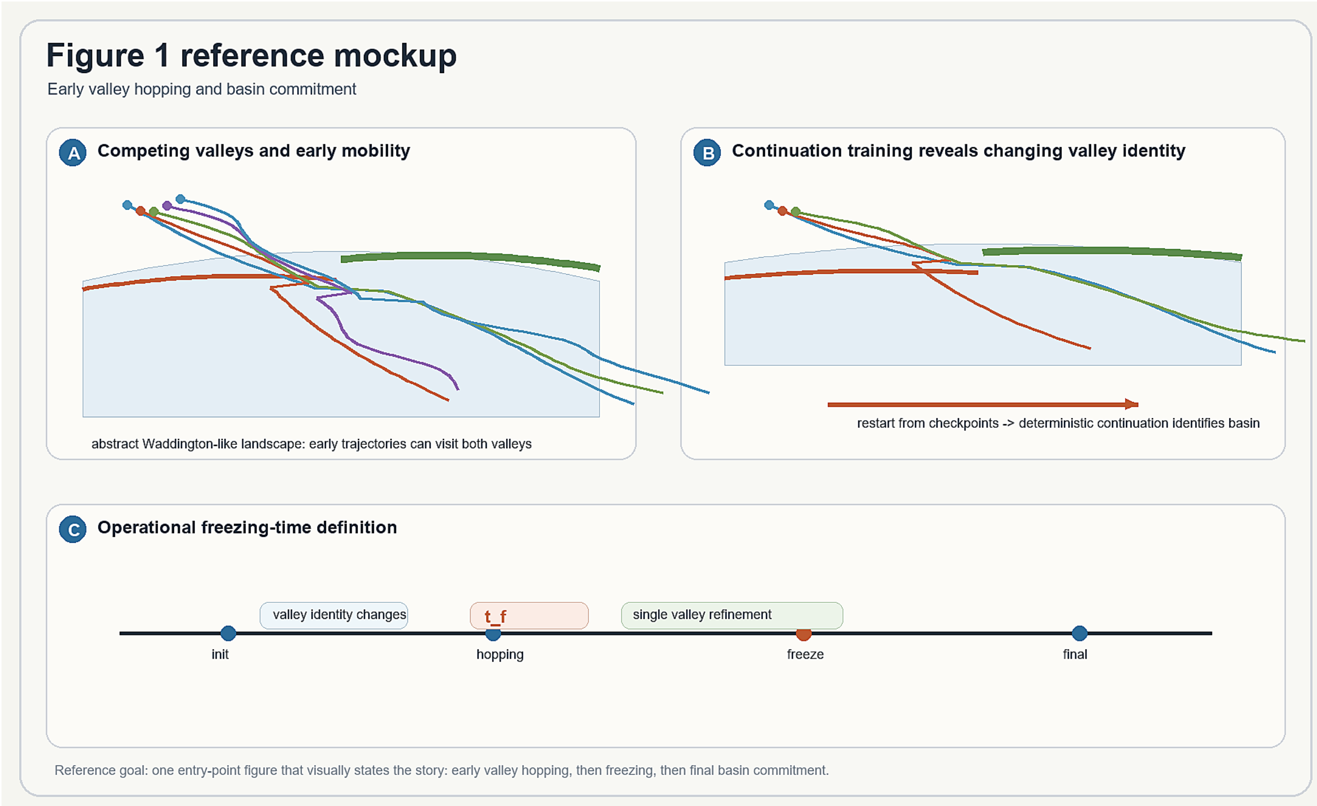

FIG. 1. Early valley hopping and basin commitment. (A) Independent SGD trajectories from the same initialization converge to multiple solution clusters, indicating competing valleys. (B) Interpolation and solution-similarity analyses show that distinct clusters are separated by loss barriers. (C) Continuation training from checkpoints t_c identifies the basin selected when stochastic exploration is stopped at that time. Early continuations end in different basin labels, whereas late continuations end in the same label. (D) The freezing time t_f is defined as the first checkpoint after which all later continuation endpoints retain the same basin label.

at which basin competition is arrested.

The final basin is set at t_f by two ingredients: the state of the landscape at that time and the valley bias present on that landscape slice. Later freezing can be beneficial because the trajectory remains mobile until after the barriers and flatness contrast have evolved further, but the empirical flatness measured at θ_{t_f} should be read as geometry at commitment rather than as a clock-independent causal effect. In this sense, the transient noisy phase is not wasted wandering; it determines which quasi-steady basin preference is locked into the final solution. The remaining task is to separate the fixed-slice valley bias from the noise-dependent stopping time at which that bias is inherited. The minimal model below is designed for this separation.

IV. A MINIMAL THEORY OF TRANSIENT FREEZING

To isolate the mechanism, we use the smallest model that contains the required ingredients. The coordinate y represents the downhill training direction. The coordinate x distinguishes two competing valleys. The valley at $x > 0$ is flatter than the valley at $x < 0$, while both valleys have equal depth relative to the separating barrier so that the selection effect is geometric rather than imposed by an energy difference. A convenient realization is

$$\mathcal{L}(x,y) = \mathcal{L}_0(y) + \begin{cases} \dfrac{x\,[x - g_1(y)]}{f_1(y)}, & x \geq 0, \\ \dfrac{x\,[x + g_2(y)]}{f_2(y)}, & x < 0, \end{cases} \tag{3}$$

where $g_{1,2}(y)$ set the valley positions and $f_{1,2}(y)$ are inverse curvatures. We denote $f_{\text{flat}}(y) \equiv f_1(y)$ and $f_{\text{sharp}}(y) \equiv f_2(y)$, with flatness contrast $\gamma \equiv f_{\text{flat}}/f_{\text{sharp}} > 1$. The barrier height between the two valleys at fixed y is $\Delta\mathcal{L}(y)$. As y increases, the valleys separate, the barrier grows, and the absolute flatness of both valleys changes. These are the empirical features needed for a freezing problem.

The dynamics follow a discrete-time Langevin update,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\left[\nabla\mathcal{L}(\boldsymbol{\theta}_t) + \boldsymbol{\xi}_t\right], \qquad \boldsymbol{\theta} = (x,y), \tag{4}$$

with zero-mean noise whose covariance mimics the

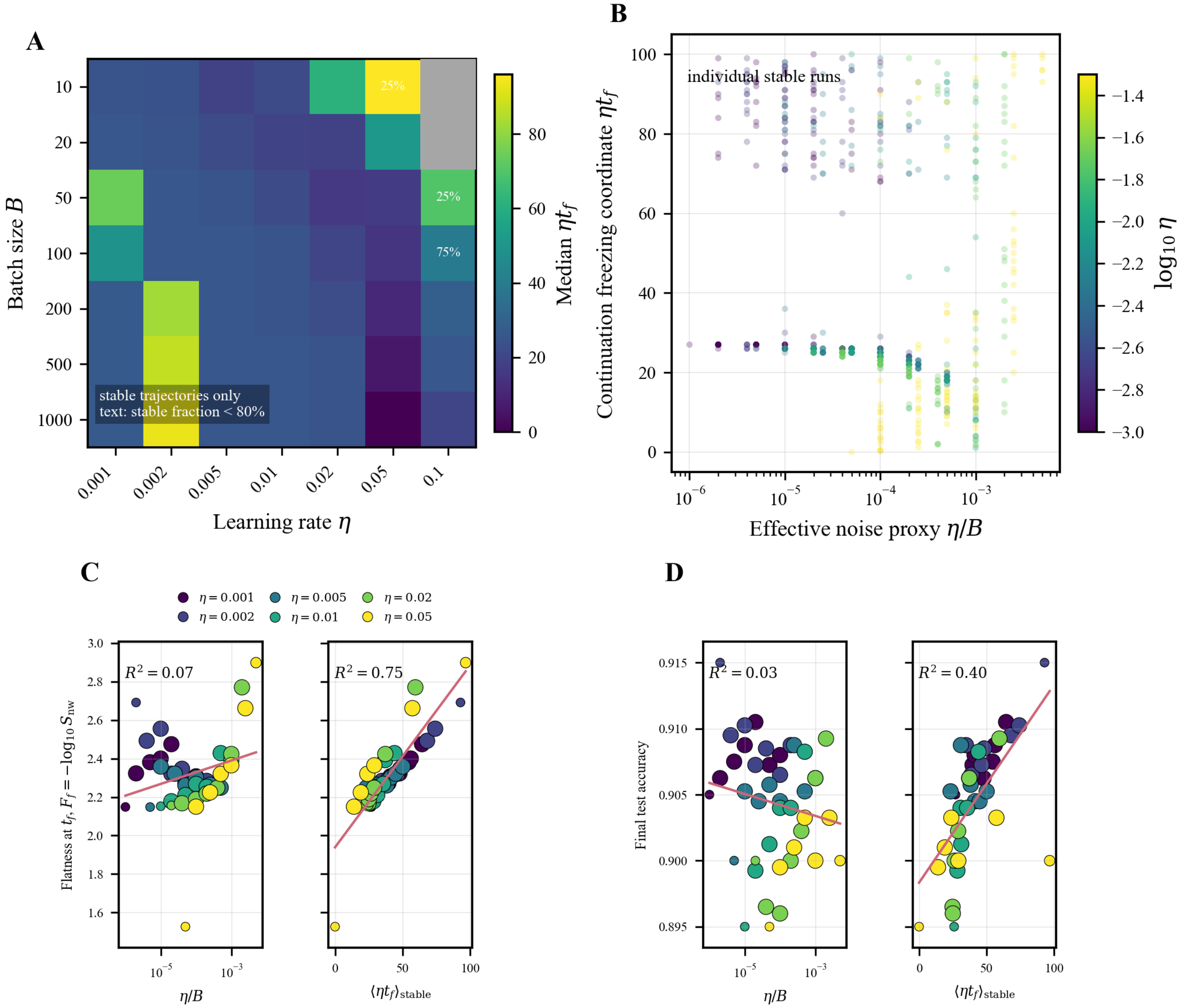

FIG. 2. Freezing time organizes basin commitment and the geometry observed at commitment. (A) Median continuation-based freezing coordinate ηt_f across batch size and learning rate; gray cells indicate conditions without final-converged stable trajectories, and text marks stable fractions below 80%. (B) Individual final-converged stable runs show the distribution of ηt_f across the effective noise proxy η/B. (C) Freezing-time flatness $F_f = -\log_{10} S_{\mathrm{nw}}(\theta_{t_f})$, measured from neuron-wise relative sharpness at the checkpoint corresponding to t_f, compared with η/B (left) and with $\langle \eta t_f \rangle$ (right). This panel reports the local geometry present when the basin label becomes stable; it is not intended as a clock-independent estimate of final flatness. (D) Final test accuracy compared with the same two organizing variables. Point colors indicate learning rate; comparison panels omit the highest-learning-rate sweep and include continuation-stable trajectories that converge at the final checkpoint.

geometry-coupled structure of SGD,

$$\mathrm{Cov}(\boldsymbol{\xi}_t) = 2\sigma\,\mathbf{H}(\boldsymbol{\theta}_t). \tag{5}$$

Here $\mathbf{H}$ is the local Hessian in the stable valley region and σ sets the minibatch noise scale. The effective noise level is $\Delta_S \sim \eta\sigma$. Thus Δ_S controls stochastic mobility, $\Delta\mathcal{L}(y)$ controls the barrier to valley exchange, and γ controls the flatness asymmetry.

The model reproduces the empirical sequence. Early in training, barriers are small and trajectories hop repeatedly between the two valleys. As y increases, escape rates decrease and hopping becomes rare. The basin is selected when the hopping time exceeds the drift time along y. Increasing Δ_S delays this crossing, keeps the system mobile for longer, and increases the probability that the final basin is the flatter valley.

The model is not a literal reduction of a neural network to two coordinates. Its role is to make the mechanism identifiable. Once competing valleys, geometry-coupled noise, and a growing barrier coexist, flat-minimum

FIG. 3. Minimal two-valley model. (A) The coordinate y describes downhill training, while x distinguishes a flat and a sharp valley. The valleys separate and the barrier grows as y increases. (B) Geometry-coupled noise is stronger along high-curvature directions, mimicking the anisotropic structure of SGD noise. (C) Simulated trajectories hop between valleys early and freeze into one valley later. (D) The model predicts that stronger effective noise Δ_S increases both the freezing time and the final probability of selecting the flat valley.

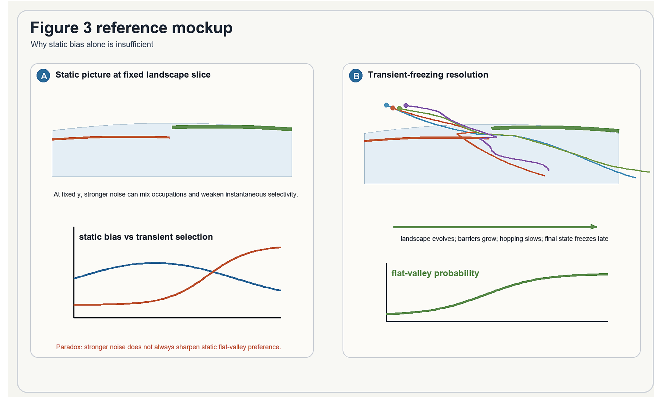

FIG. 4. Static bias and its limitation. (A) On a fixed landscape slice, anisotropic noise induces an effective loss that favors the flatter valley. (B) The corresponding quasi-steady flat-valley probability $P_{\text{flat}}^{\text{ss}}(y)$ is larger than the equilibrium expectation. (C) At fixed y, increasing Δ_S can reduce instantaneous selectivity by mixing the two valleys. (D) The apparent contradiction is resolved by recognizing that stronger noise delays freezing, so the final probability is inherited from a later landscape slice.

selection becomes a history-dependent consequence of nonequilibrium dynamics rather than only a property of a fixed landscape.

V. STATIC BIAS IS REAL BUT INSUFFICIENT

The reduced model also explains why a fixed-landscape theory is incomplete. At an approximately fixed y, the transverse coordinate x can relax faster than the longitudinal drift. In this quasi-steady regime, anisotropic SGD noise generates a nonequilibrium preference for the flatter valley [11, 12, 14]. Let $\delta f(y) \equiv f_{\text{sharp}}(y) - f_{\text{flat}}(y)$. Kramers-rate estimates give the flat-valley occupation probability

$$P_{\text{flat}}^{\text{ss}}(y) \approx \left[1 + \gamma^{-\frac{1}{2}} \exp\left(\frac{\Delta\mathcal{L}(y)\delta f(y)}{2\Delta_S}\right)\right]^{-1}. \tag{6}$$

Because $f_{\text{flat}} > f_{\text{sharp}}$, $\delta f < 0$ and this expression is biased toward the broader valley. Equivalently, the anisotropic noise reshapes the effective landscape by lowering the flat valley relative to the sharp one.

However, Eq. (6) also exposes the limitation of a static picture. If y is held fixed, increasing Δ_S does not necessarily make the flat valley more selectively occupied. A larger effective noise level can mix the two valleys and weaken the instantaneous fixed-slice preference. This trend is opposite to the observed final outcome, where stronger SGD noise often leads to a higher probability of ending in flatter minima.

The resolution is that the final outcome is not a steady-state probability evaluated at an arbitrary fixed y. The landscape evolves while the escape rates collapse. The probability that matters is the quasi-steady bias at the last point where inter-valley hopping remains effective. Stronger noise can therefore reduce the fixed-slice selectivity and still enhance final flat-minimum selection, because it moves the freezing point to a later part of the landscape.

This distinction is the core of the transient-freezing mechanism. Static bias determines the direction of selection on a given landscape slice. Freezing determines which slice is inherited as the final basin choice. Both ingredients are needed to explain why noise promotes flat minima during real training.

VI. QUANTITATIVE PREDICTIONS AND EXPERIMENTAL TESTS

The theory becomes predictive when the final occupation is written as a stopping-time quantity. Let y_f be the training coordinate at which inter-valley hopping becomes negligible. Then

$$P_{\text{flat}}^{\text{tr}} \approx P_{\text{flat}}^{\text{ss}}(y_f)\,. \tag{7}$$

This expression does not assume that training reaches a global steady state. It assumes only that, before freezing, the transverse valley coordinate relaxes faster than the downhill coordinate, so the last quasi-steady bias is inherited by the final basin.

The freezing coordinate is determined by a competition of time scales. The inter-valley escape time is $\tau_{\text{esc}} \sim (k_{f\to s} + k_{s\to f})^{-1}$, whereas the landscape changes on a drift time $\tau_y \sim |\dot{y}|^{-1}$. Freezing occurs when τ_{esc} becomes longer than the time over which the quasi-steady distribution changes appreciably. In the two-valley model this criterion gives, up to slowly varying prefactors,

$$y_f \approx y_b \frac{\sqrt{\Delta_S \ln\left(\Delta_S/\varepsilon^2\Phi^2\right)}}{x_2 - \sqrt{\Delta_S \ln\left(\Delta_S/\varepsilon^2\Phi^2\right)}}, \tag{8}$$

where ε defines the arrest criterion and Φ collects model parameters that vary slowly compared with the escape rates. The important consequence is monotonic: larger Δ_S pushes the freezing point to larger y over the relevant range.

Substituting this freezing point into Eq. (7) yields a final flat-valley probability that increases with Δ_S for $\gamma > 1$,

$$P_{\mathrm{flat}}^{\mathrm{tr}} \approx \left[1 + \gamma^{-\frac{1}{2}} \left(\frac{\sqrt{\Delta_S}}{\varepsilon\Phi}\right)^{1-\gamma}\right]^{-1}. \tag{9}$$

Thus the model predicts the empirical trend that stronger noise promotes flat-minimum selection, even though the fixed-slice selectivity alone would suggest the opposite.

The framework makes three direct predictions. First, increasing SGD noise increases the freezing time over the stable training range. Second, the geometry observed at the freezing checkpoint and the associated generalization outcomes are organized more directly by t_f than by raw hyperparameters alone. Third, different optimizers or schedules become comparable when they generate the same ordering of time scales: early inter-valley relaxation followed by barrier-driven arrest. These predictions can be tested by plotting freezing-time flatness, test performance, and flat-valley probability against t_f across learning rates and batch sizes.

VII. DISCUSSION

We have shown that flat-minimum selection in SGD can be viewed as a nonequilibrium freezing problem. The central mechanism is that SGD first remains mobile among competing valleys and then becomes committed when inter-valley motion stops. The final basin is therefore selected at a finite stopping time, not only by a late-time steady state inside an already chosen valley.

This perspective clarifies the role of noise. SGD noise is not merely a scalar regularizer that always increases an instantaneous preference for flatness. Its main role is dynamical: it controls the duration of the exploratory window before basin commitment. Stronger noise can keep the trajectory mobile until later in training, where the evolving landscape and the geometry-induced bias together make selection of the flatter valley more likely.

The mechanism also gives a design principle. Learning-rate and batch-size schedules should be understood as tools for controlling the timing of basin commitment, not only as tools for guaranteeing convergence. A schedule that preserves inter-valley mobility too briefly may freeze into a sharp basin; a schedule that preserves it too long may prevent reliable convergence. The useful objective is to regulate the freezing time so that exploration persists until favorable valleys are accessible and then arrests before late-time noise destroys refinement.

The model is deliberately minimal, so its scope conditions are explicit. The transient-freezing picture should apply when there are multiple competing valleys, geometry-coupled stochasticity, barriers that evolve during descent, and a finite time window in which valley exchange is possible. These ingredients are not specific to the small network used here and are expected in richer architectures and datasets. What may change across systems is the most informative continuation protocol, endpoint-similarity measure, and dimensionality of the effective valley-coordinate space.

More broadly, the results connect the implicit bias of SGD to a class of nonequilibrium selection problems in which the outcome depends on the pathway to arrest. Fixed-landscape effective potentials remain useful because they determine the local direction of the bias. They become predictive of the final solution only when combined with the freezing time at which that bias is locked in. From this viewpoint, SGD selects flat minima not simply because they are statically favored, but because noise delays the freezing point at which basin competition is resolved.

DATA AND CODE AVAILABILITY

The code used for the empirical experiments and the two-valley simulations is available at `https://github.com/YangNing1995/Transient_Dynamics_SGD`.

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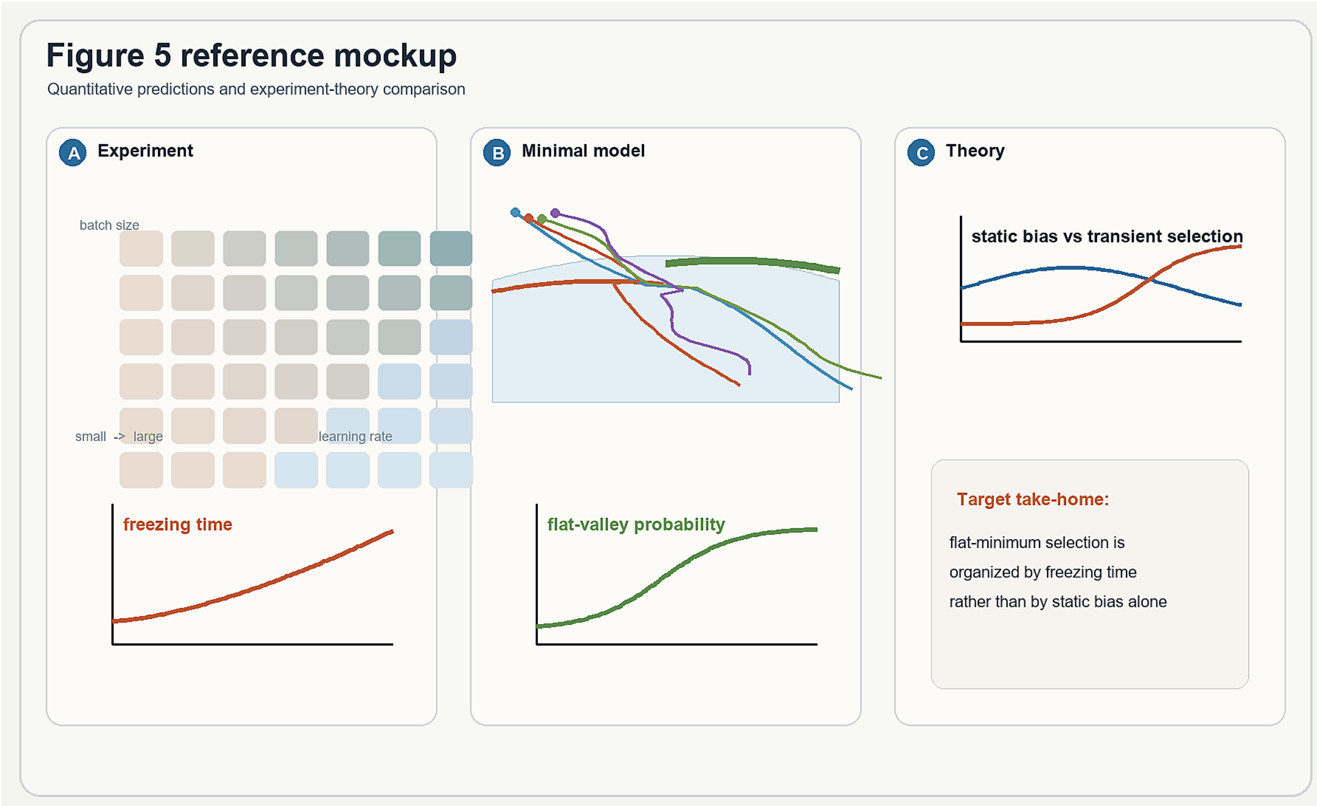

FIG. 5. Predictions and experiment-theory comparison. (A) The transient theory predicts that stronger effective noise delays freezing. (B) Evaluating the quasi-steady flat-valley bias at the freezing point predicts an increasing final flat-valley probability with noise. (C) Minimal-model simulations test the predicted relation between Δ_S, t_f, and P_{flat}. (D) Neural-network data are organized by the same variables: freezing-time flatness and test performance plotted against the measured freezing time. This comparison tests whether t_f is the stopping-time variable that organizes basin-commitment geometry and the inherited outcome.

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